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Redfish-native Fleet-Scale GPU Observability

for AI Cluster Operational Efficiency

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AGENDA

This talk will cover

WHY

- GPU cloud operational reality
- Efficiency at AI scale runs on observability & digital twins

HOW

- Why Redfish is the right substrate for the twin
- RCA use-case demos

WHAT NEXT

- Opportunities & call to action
- Q&A & discussion

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GPU CLOUD OPERATIONAL REALITY

Operational excellence is the neocloud's product

For AI tenants, how a GPU cloud handles failure is the product. Leading GPU clouds differentiate through operational workflows tenants experience directly, supported by observability across hardware and workloads.

Communicate Clearly

What's happening, the RCA steps underway, and an ETA for the fix.

Remediate quickly

For isolated node failures, restore or replace capacity quickly instead of blocking recovery on troubleshooting.

Own outages

Take responsibility for lost time.

Why the bar is this high

Training — Synchronized jobs can stall when a participating GPU or NIC fails, unless resilience mechanisms isolate and recover the fault.

Inference — Serving is SLA-bound: a degraded GPU breaks a sharded replica and its latency SLOs, so staying in SLA means detecting and draining bad hardware fast.

The enabler — Facility operators and neoclouds own the physical-infrastructure layers, while tenants own workload health, with some responsibilities shared.

Operational patterns per SemiAnalysis ClusterMAX™ 2.0 (independent GPU-cloud rating; 140+ tenant interviews, Nov 2025).

THE THESIS

Efficiency at AI scale will run on digital twins

AI-cloud operators are increasingly exploring continuously updated fleet models, or digital twins, fed by standards-based APIs.

Monitoring

Is it up? Threshold alerts.

Fleet observability

Why is it behaving this way? OOB telemetry + shareable evidence.

Digital twin

A software-defined, queryable model that drives efficiency, RCA & tenant SLAs.

The twin is only as good as the data plane beneath it — which is where Redfish comes in.

CENTERPIECE

Two twins, one data plane

Two twins, different scopes, owners and access models — mirroring the ownership stack: physical infrastructure (facility + rack) vs. the workload.

INFRASTRUCTURE TWIN

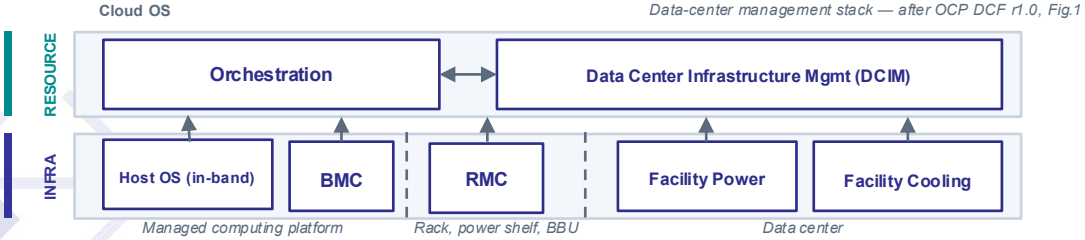
Manage the plant

- Scope: facility, data hall, rack, server
- Power & cooling (CDU), rack PDUs, node health
- Owners: colo facility ops + neocloud
- Data: OOB physical telemetry, config state

AI-CLUSTER-AWARE TWIN

Manage the workload

- Scope: grouped by workload type / tenant
- Workload health, SLA & efficiency
- Read rights: hardware telemetry (scoped)
- Tenant-facing — often shared with the tenant



Shared data plane: one vendor-neutral Redfish model feeds both twins — same telemetry, scoped differently.

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WHY REDFISH

The right substrate for a twin — in a human- and machine-friendly language

Aggregation & proxy

The whole stack — facility DCIM (power/cooling/CDU) to rack to server BMC — through a common model and aggregated management view.

Adoption

Adopted by hardware vendors across the stack.

Requirements as Interop Profiles

Machine-checkable conformance you can hand your provider as your requirement — the OCP Hardware Management workgroup has published good examples.

Secure and open

Open DMTF stack from interface to device layer; role-based access + SPDM measurements and attestation — supporting stronger verification of device identity and measurements

One vendor-neutral schema

A single model across an OEM/ODM fleet.

Relevant and continuously evolving

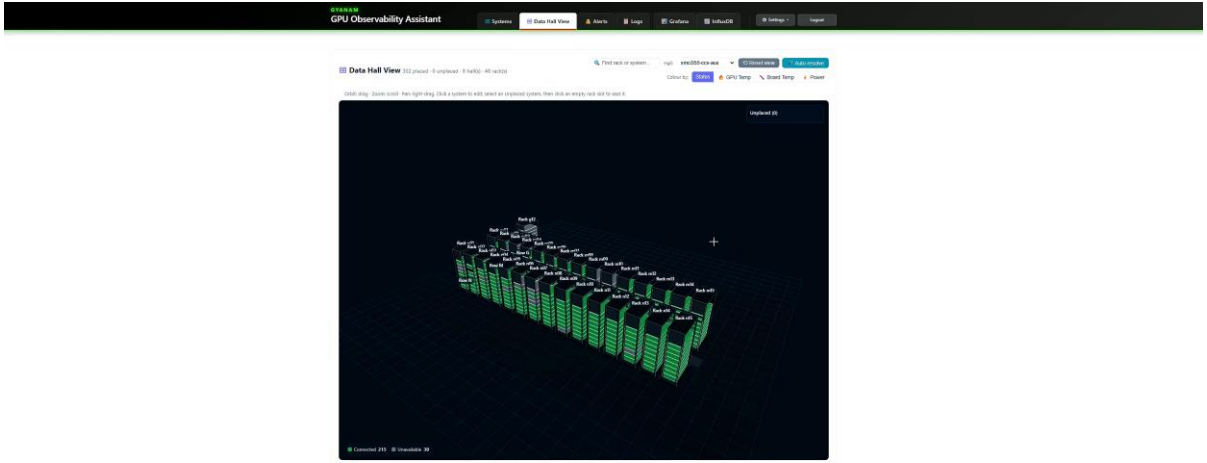
Keeps pace with modern DCIM needs — quarterly releases, well supported by the member companies.

USE CASE 1 · THERMAL

Fleet-wide thermal outlier triage

Workloads are running slow or getting throttled due to thermal conditions — and you need to see it across the fleet.

1. **Fleet heatmap surfaces hot outliers**
2. **Drill into the per-system view**
3. **Export the evidence bundle**



- Feeds both the infrastructure twin and the AI-cluster-aware twin
- Can trigger RCA/triage to the liquid-cooling / heat-sink vendor

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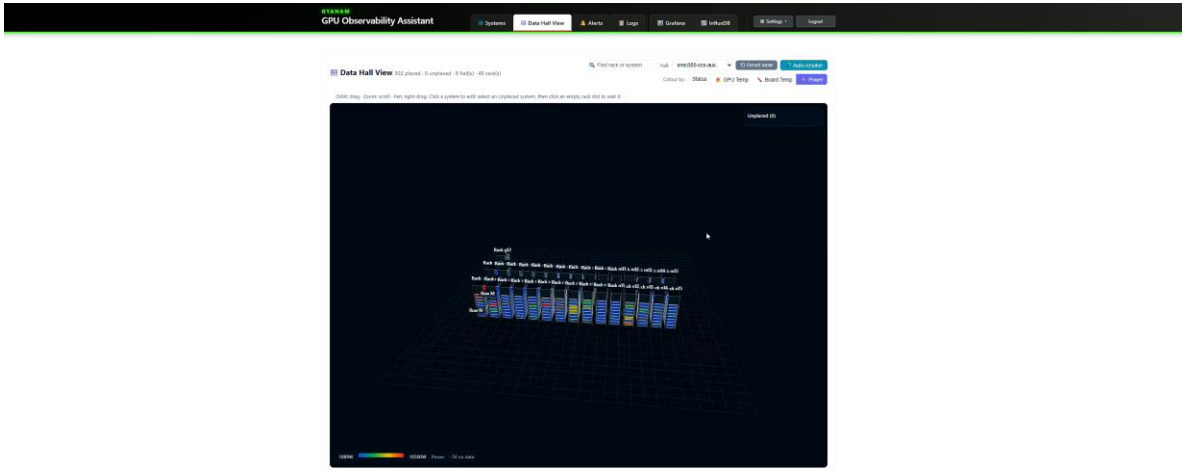
USE CASE 2 · POWER FAULTS

Transient or permanent power anomalies

The mystery crash that leaves no trace — in training or inference. One possible cause is a workload-induced power anomaly.

1. **Per-rail VR voltage / current / temperature**
2. **Spot the drifting rail vs fleet baseline**
3. **Identify drift that may precede a hard fault**

- FA engineer needs RCA packet



Opportunity — *Observed rail-telemetry implementations commonly sample at approximately 1–10 Hz reflecting management-overhead constraints, sufficient for slow drift but not sub-cycle transients. The opportunity is on-demand, higher-frequency telemetry for targeted investigation.*

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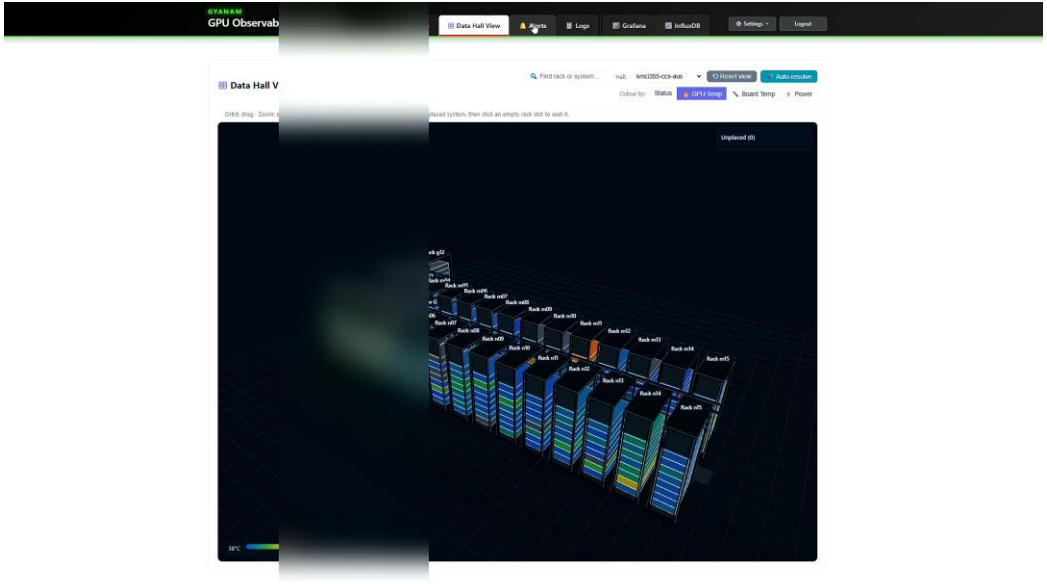
USE CASE 3 · SPORADIC FAULT

Sporadic fault to reproducible RCA evidence

The challenge is a sporadic fault that may disappear before an engineer can begin investigating it. The goal is to capture the right evidence while the failure context is still available.

1. **Redfish event triggers on anomaly**
2. **Diagnostic bundle collected at the node**
3. **Consistent export captures the window**
4. **RCA package**

- Customer-initiated RCA evidence sharing



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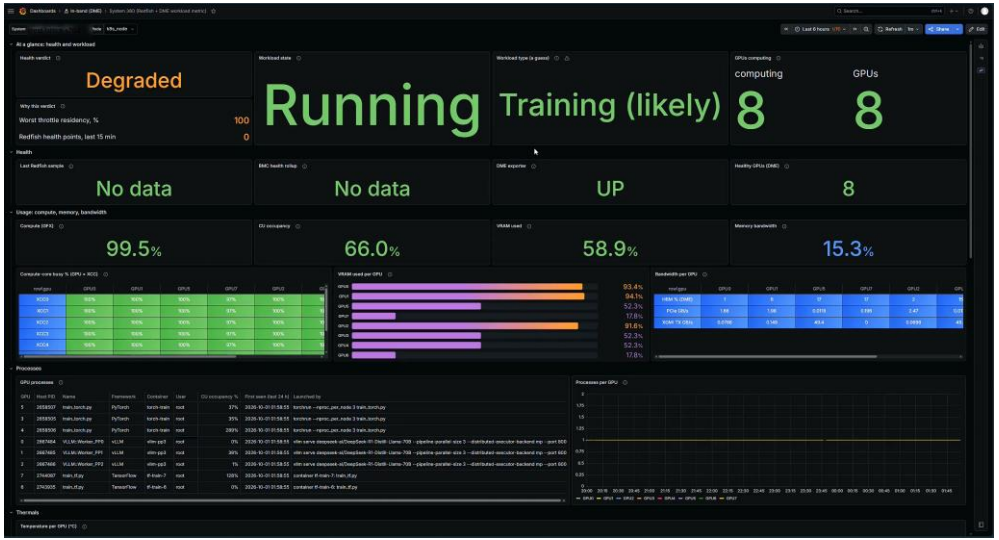
USE CASE 4 · WORKLOAD INDUCED RCA

Workload-induced behavior RCA

Most valuable during early model & workload bring-up: correlate fleet telemetry to attribute a crash — and harden the full stack.

1. **Bound the behavior to the node group**
2. **Correlate telemetry across the window**
3. **Assess evidence: hardware, workload, or interaction**

- The AI-cluster-aware twin in action
- R&D feedback: developers harden the full stack



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TIE IT TOGETHER

Use cases mapped to the two twins and vendor RCA loop

Use case	Infra twin	AI-cluster-aware twin	Vendor RCA loop
Thermal outlier triage	✓	✓	✓
Power anomalies	✓	✓	✓
Sporadic fault RCA		✓	✓
Workload-induced RCA		✓	✓

Together, these use cases illustrate the data and workflows required by both twins, and the vendor RCA loop between them.

Vendor RCA loop = root-cause-analysis feedback to the responsible vendor: GPU / board for silicon & power faults, or the liquid-cooling / heat-sink vendor for thermal.

OPPORTUNITIES

Where the workflow excels — and additional opportunities

Tiered evidence control plane (WIP)

Selective collection, matched to the operational tier.

Tier 1 · default — SEL, RAS as Redfish Events; periodic default telemetry, (repeats suppressed metrics).

Tier 2 — higher-rate metric reports on demand; on-change triggers + policy service definitions + diagnostic-log collection.

Tier 3 · opportunity — per-device crash dump + on-demand bulk telemetry (CollectDiagnosticData / CollectTelemetryData).

Tenant-scoped metrics

In-band export spends host CPU + GPU-mgmt bandwidth; out-of-band collection stays preferred.

Same service, in-band — the Redfish Host Interface (DSP0270) reaches the same service locally: same schemas & calls.

Opportunity — so the gap is authorization: a Tenant role with scoped read privileges for tiered telemetry collection (DSP0270 bootstrap can mint tenant user and scope).

Vendor-independent naming (WIP)

Redfish is hypermedia: Identifiers are named per vendor, discovered by clients. — hard across a mixed fleet.

Opportunity — Redfish aliasing and alias-based instance labeling give vendor-neutral metric names.

Redfish-native reference twin

The Redfish validators, emulator, mockups, compliance suites and Interop Profiles already lowered the cost of adopting Redfish.

Opportunity — an open, vendor-neutral, Redfish-native reference observability twin familiarizes the concept and adoption.

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WHERE THIS IS MOVING

Each opportunity builds on published capabilities and active work

Tiered evidence control plane

Redfish 2025.2 introduced the TelemetryData resource supporting bulk telemetry workflows through CollectTelemetryData — the open work is standardizing the payloads.

-> In flight/WIP: PLDM DSP2091 Bulk Telemetry, DSP2092 Crash Log, DSP2093 PLDM WDT modeling.

Tenant scoping

Roles & Privileges scope access; the Redfish Host Interface (DSP0270) reaches the same redfish service in-band — so the gap is authorization, not transport.

-> a first-class Tenant role with resource-scoped read privileges for tiered telemetry access — a proposal, not yet chartered.

Vendor-independent metric naming

The Redfish Forum Telemetry Task Force charter: identify a metric instance without relying on vendor-specific Id values.

-> In flight/WIP: Alias and read-only ServiceAlias, Name/Label construction rules.

Redfish-native reference twin

Conformance tools, test suites and Interop Profiles already lowered Redfish's cost of adoption.

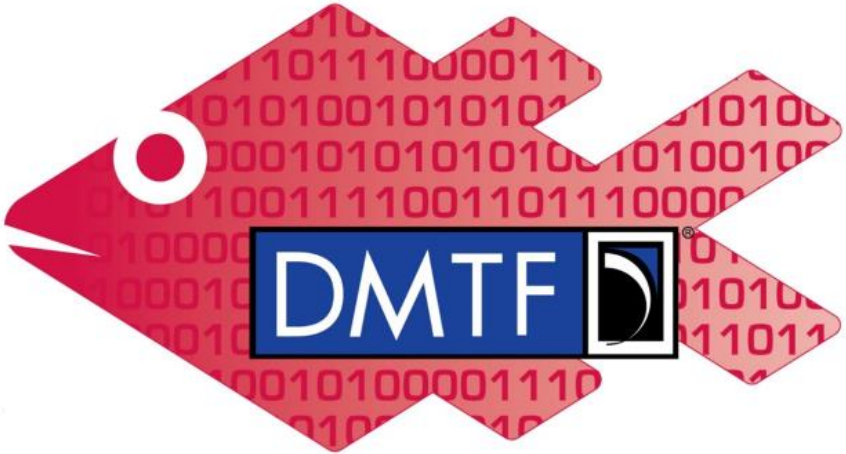
-> an open, vendor-neutral, Redfish-native reference observability twin. — a proposal, not yet chartered.

Call to Action

- **Engage —** engage with the DMTF Redfish Forum and relevant working groups through the appropriate participation channels.
- **Collaborate —** AI companies & GPU tenants: share your neocloud requirements and challenges. Neoclouds: share your vendor requirements.
- **Adopt —** pilot Redfish solutions in your data centers.
- **Partner —** work with the standards bodies to shape future enhancements.

Fleet-scale GPU observability is the foundation for both digital twins.

Q&A & Discussion



Redfish